

# AERIS Expert Review Packet

## Anomaly

|                              |                                                                |
|------------------------------|----------------------------------------------------------------|
| Source / metric              | tceq / Nitrogen dioxide (no2)                                  |
| Observed / expected baseline | 28.8 / 4.73172 ppb                                             |
| Baseline method              | mean of the preceding 7 days of this series (Z-score detector) |
| Time                         | 2026-07-08 00:00 UTC                                           |
| Location                     | 29.8342, -95.4891                                              |
| Severity                     | severe                                                         |
| Detectors                    | zscore, stl, isolation_forest                                  |

## Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

| Source   | Metric                                       | Unit    | Entities / points | Range; mean                      | Nearest to event                               |
|----------|----------------------------------------------|---------|-------------------|----------------------------------|------------------------------------------------|
| asos     | Relative humidity (humidity)                 | percent | 9 / 839           | 39.12 to 100; mean 73.707        | 55.64 at 2026-07-07 23:55Z; 5 min before event |
| asos     | Air temperature (temperature)                | degC    | 9 / 839           | 21.6667 to 36.1111; mean 28.8204 | 32 at 2026-07-07 23:55Z; 5 min before event    |
| asos     | Wind direction (wind_direction)              | degree  | 9 / 1094          | 0 to 360; mean 134.232           | 0 at 2026-07-07 23:55Z; 5 min before event     |
| asos     | Wind speed (wind_speed)                      | m/s     | 9 / 1126          | 0 to 15.4333; mean 2.2433        | 0 at 2026-07-07 23:55Z; 5 min before event     |
| noaa_gfs | 500-hPa geopotential height (gh_500)         | m       | 10 / 130          | 5891.13 to 5924.31; mean 5909.1  | 5911.33 at 2026-07-08 00:00Z; at event time    |
| noaa_gfs | Planetary boundary layer height (pbl_height) | m       | 10 / 130          | 19.6394 to 2239.24; mean 721.26  | 1282.97 at 2026-07-08 00:00Z; at event time    |
| noaa_gfs | Precipitable water (precipitable_water)      | mm      | 10 / 130          | 37.6598 to 52.9161; mean 45.7319 | 49.6827 at 2026-07-08 00:00Z; at event time    |
| noaa_gfs | Surface pressure (surface_pressure)          | hPa     | 10 / 130          | 1006.52 to 1017.31; mean 1013.23 | 1012.12 at 2026-07-08 00:00Z; at event time    |
| noaa_gfs | 850-hPa temperature (t_850)                  | degC    | 10 / 130          | 17.4065 to 20.4926; mean 19.0969 | 18.4321 at 2026-07-08 00:00Z; at event time    |
| noaa_gfs | 10-m eastward wind (u_10m)                   | m/s     | 10 / 130          | -2.83889 to 1.82714; mean 0.3481 | -0.758887 at 2026-07-08 00:00Z; at event time  |
| noaa_gfs | 10-m northward wind (v_10m)                  | m/s     | 10 / 130          | -0.854414 to 4.9494; mean 2.1456 | 3.8694 at 2026-07-08 00:00Z; at event time     |
| openaq   | Ozone (ozone)                                | ppm     | 17 / 785          | -0.004 to 0.069; mean 0.0172     | 0.015 at 2026-07-08 00:00Z; at event time      |

| Source      | Metric                                                | Unit    | Entities / points | Range; mean                             | Nearest to event                                          |
|-------------|-------------------------------------------------------|---------|-------------------|-----------------------------------------|-----------------------------------------------------------|
| openaq      | PM10 (pm10)                                           | ug/m3   | 3 / 150           | 8 to 75; mean 25.5467                   | 41 at 2026-07-08 00:00Z; at event time                    |
| openaq      | PM2.5 (pm25)                                          | ug/m3   | 79 / 2718         | 0 to 62.1; mean 5.5521                  | 12.6 at 2026-07-08 00:00Z; at event time                  |
| openweather | Cloud cover (cloud_cover)                             | percent | 5 / 360           | 0 to 100; mean 26.8861                  | 43 at 2026-07-08 00:01Z; 2 min after event                |
| openweather | Relative humidity (humidity)                          | percent | 5 / 360           | 38 to 95; mean 75.7333                  | 63 at 2026-07-08 00:01Z; 2 min after event                |
| openweather | Precipitation rate (precipitation)                    | mm/h    | 5 / 360           | 0 to 5.62; mean 0.0476                  | 0 at 2026-07-08 00:01Z; 2 min after event                 |
| openweather | Surface pressure (pressure)                           | hPa     | 5 / 360           | 1012 to 1019; mean 1016.26              | 1014 at 2026-07-08 00:01Z; 2 min after event              |
| openweather | Air temperature (temperature)                         | degC    | 5 / 360           | 23.78 to 36.05; mean 29.5481            | 32.03 at 2026-07-08 00:01Z; 2 min after event             |
| openweather | Wind direction (wind_direction)                       | degree  | 5 / 360           | 0 to 359; mean 190.558                  | 111 at 2026-07-08 00:01Z; 2 min after event               |
| openweather | Wind speed (wind_speed)                               | m/s     | 5 / 360           | 0.36 to 8.39; mean 2.2549               | 0.89 at 2026-07-08 00:01Z; 2 min after event              |
| purpleair   | PM2.5 (pm25)                                          | ug/m3   | 74 / 5125         | 0 to 146.4; mean 7.3305                 | 18.7 at 2026-07-08 00:00Z; at event time                  |
| sentinel5p  | Aerosol index granules (s5p_aer_ai_granule_available) | count   | 8 / 8             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| sentinel5p  | CH4 granules available (s5p_ch4_granule_available)    | count   | 8 / 8             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| sentinel5p  | CO column (s5p_co_column)                             | mol/m^2 | 6 / 6             | 0.0243723 to 0.0253077; mean 0.0249     | 0.0249873 at 2026-07-07 18:57Z; 5 h 2 min before event    |
| sentinel5p  | CO granules available (s5p_co_granule_available)      | count   | 8 / 8             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| sentinel5p  | HCHO column (s5p_hcho_column)                         | mol/m^2 | 8 / 8             | 0.000174097 to 0.000235341; mean 0.0002 | 0.000202795 at 2026-07-07 20:37Z; 3 h 22 min before event |
| sentinel5p  | HCHO granules available (s5p_hcho_granule_available)  | count   | 8 / 8             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| sentinel5p  | NO2 column (s5p_no2_column)                           | mol/m^2 | 4 / 4             | 3.63106e-05 to 4.30727e-05; mean 0      | 4.10314e-05 at 2026-07-07 20:37Z; 3 h 22 min before event |
| sentinel5p  | NO2 granules available (s5p_no2_granule_available)    | count   | 4 / 4             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| sentinel5p  | O3 granules available (s5p_o3_granule_available)      | count   | 8 / 8             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| sentinel5p  | SO2 column (s5p_so2_column)                           | mol/m^2 | 8 / 8             | -3.49304e-05 to 4.72775e-05; mean 0     | 2.51087e-05 at 2026-07-07 20:37Z; 3 h 22 min before event |
| sentinel5p  | SO2 granules available (s5p_so2_granule_available)    | count   | 8 / 8             | 1 to 1; mean 1                          | 1 at 2026-07-07 20:37Z; 3 h 22 min before event           |
| tceq        | Carbon monoxide (co)                                  | ppm     | 3 / 162           | 0 to 1; mean 0.237                      | 0.6 at 2026-07-08 00:00Z; at event time                   |
| tceq        | Nitrogen dioxide (no2)                                | ppb     | 12 / 637          | 0 to 35.8; mean 6.5196                  | 28.8 at 2026-07-08 00:00Z; at event time                  |

| Source | Metric               | Unit | Entities / points | Range; mean               | Nearest to event                         |
|--------|----------------------|------|-------------------|---------------------------|------------------------------------------|
| tceq   | Sulfur dioxide (so2) | ppb  | 3 / 154           | -0.5 to 6.8; mean -0.0481 | -0.4 at 2026-07-08 00:00Z; at event time |

## Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

| Source      | Metric                                       | Values                                                                                                                                                                             |
|-------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asos        | Relative humidity (humidity)                 | 07-06 12Z 70.19, 18Z 55.49, 07-07 00Z 77.29, 06Z 89.88, 12Z 71, 18Z 57, 07-08 00Z 79.08, 06Z 92.24, 12Z 70.83, 18Z 59.59, 07-09 00Z 77.1, 06Z 89.2                                 |
| asos        | Air temperature (temperature)                | 07-06 12Z 30.32, 18Z 32.62, 07-07 00Z 28.31, 06Z 25.72, 12Z 29.83, 18Z 32.19, 07-08 00Z 25.82, 06Z 24.3, 12Z 30.32, 18Z 31.65, 07-09 00Z 28.27, 06Z 25.68                          |
| asos        | Wind direction (wind_direction)              | 07-06 12Z 180.9, 18Z 166.7, 07-07 00Z 137.1, 06Z 73.4, 12Z 133, 18Z 148.1, 07-08 00Z 105.8, 06Z 27.45, 12Z 187.1, 18Z 171.8, 07-09 00Z 158.6, 06Z 126.2                            |
| asos        | Wind speed (wind_speed)                      | 07-06 12Z 2.058, 18Z 3.818, 07-07 00Z 2.083, 06Z 0.7665, 12Z 1.424, 18Z 3.304, 07-08 00Z 2.915, 06Z 0.3517, 12Z 2.173, 18Z 3.988, 07-09 00Z 2.757, 06Z 1.442                       |
| noaa_gfs    | 500-hPa geopotential height (gh_500)         | 07-06 12Z 5892, 18Z 5902, 07-07 00Z 5898, 06Z 5905, 12Z 5899, 18Z 5914, 07-08 00Z 5912, 06Z 5918, 12Z 5911, 18Z 5924, 07-09 00Z 5915, 06Z 5919, 12Z 5910                           |
| noaa_gfs    | Planetary boundary layer height (pbl_height) | 07-06 12Z 175.4, 18Z 1810, 07-07 00Z 1019, 06Z 227.9, 12Z 126.9, 18Z 1996, 07-08 00Z 976.4, 06Z 250.5, 12Z 150.5, 18Z 1936, 07-09 00Z 244.1, 06Z 303.7, 12Z 159.8                  |
| noaa_gfs    | Precipitable water (precipitable_water)      | 07-06 12Z 41.89, 18Z 42.78, 07-07 00Z 46.68, 06Z 45.56, 12Z 46.22, 18Z 48.1, 07-08 00Z 48.87, 06Z 47.17, 12Z 46.64, 18Z 48.04, 07-09 00Z 45.44, 06Z 43.81, 12Z 43.31               |
| noaa_gfs    | Surface pressure (surface_pressure)          | 07-06 12Z 1012, 18Z 1012, 07-07 00Z 1010, 06Z 1013, 12Z 1014, 18Z 1014, 07-08 00Z 1012, 06Z 1015, 12Z 1015, 18Z 1015, 07-09 00Z 1013, 06Z 1014, 12Z 1014                           |
| noaa_gfs    | 850-hPa temperature (t_850)                  | 07-06 12Z 19.67, 18Z 17.97, 07-07 00Z 19.75, 06Z 19.78, 12Z 18.77, 18Z 17.85, 07-08 00Z 19.03, 06Z 19.77, 12Z 19.06, 18Z 17.99, 07-09 00Z 19.87, 06Z 19.53, 12Z 19.2               |
| noaa_gfs    | 10-m eastward wind (u_10m)                   | 07-06 12Z 1.021, 18Z 0.7598, 07-07 00Z -0.6914, 06Z 0.346, 12Z 0.8611, 18Z -0.1427, 07-08 00Z -1.34, 06Z 0.2456, 12Z 0.8352, 18Z 0.9613, 07-09 00Z -0.07786, 06Z 0.7396, 12Z 1.007 |
| noaa_gfs    | 10-m northward wind (v_10m)                  | 07-06 12Z 1.384, 18Z 1.541, 07-07 00Z 3.52, 06Z 2.189, 12Z 1.199, 18Z 1.606, 07-08 00Z 4.025, 06Z 2.566, 12Z 1.287, 18Z 1.112, 07-09 00Z 3.109, 06Z 2.865, 12Z 1.49                |
| openaq      | Ozone (ozone)                                | 07-06 12Z 0.01262, 18Z 0.03414, 07-07 00Z 0.01345, 06Z 0.005189, 12Z 0.01275, 18Z 0.03023, 07-08 00Z 0.02554, 06Z 0.004395, 12Z 0.004094, 18Z 0.03667                              |
| openaq      | PM10 (pm10)                                  | 07-06 12Z 30.22, 18Z 27.5, 07-07 00Z 15.89, 06Z 18.28, 12Z 33.17, 18Z 29.88, 07-08 00Z 21.94, 06Z 24.39, 12Z 36, 18Z 23                                                            |
| openaq      | PM2.5 (pm25)                                 | 07-06 12Z 6.452, 18Z 5.623, 07-07 00Z 4.88, 06Z 6.191, 12Z 6.682, 18Z 7.206, 07-08 00Z 4.586, 06Z 6.593, 12Z 5.806, 18Z 3.239, 07-09 00Z 3.701, 06Z 4.591, 12Z 4.524               |
| openweather | Cloud cover (cloud_cover)                    | 07-06 12Z 24.47, 18Z 24.9, 07-07 00Z 14.61, 06Z 4.033, 12Z 36.77, 18Z 46.57, 07-08 00Z 27.8, 06Z 4.667, 12Z 6.655, 18Z 29.1, 07-09 00Z 69.65, 06Z 31.8                             |
| openweather | Relative humidity (humidity)                 | 07-06 12Z 78.2, 18Z 59.53, 07-07 00Z 75.58, 06Z 87.8, 12Z 77.07, 18Z 59.23, 07-08 00Z 76.6, 06Z 90.5, 12Z 79.59, 18Z 60.79, 07-09 00Z 75.35, 06Z 88.2                              |
| openweather | Precipitation rate (precipitation)           | 07-06 12Z 0, 18Z 0.005333, 07-07 00Z 0, 06Z 0, 12Z 0, 18Z 0, 07-08 00Z 0.5237, 06Z 0, 12Z 0, 18Z 0.04379, 07-09 00Z 0, 06Z 0                                                       |
| openweather | Surface pressure (pressure)                  | 07-06 12Z 1016, 18Z 1014, 07-07 00Z 1014, 06Z 1016, 12Z 1018, 18Z 1016, 07-08 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1017, 07-09 00Z 1016, 06Z 1016                                     |
| openweather | Air temperature (temperature)                | 07-06 12Z 29.63, 18Z 34.39, 07-07 00Z 29.5, 06Z 26.39, 12Z 29.93, 18Z 34.09, 07-08 00Z 27.7, 06Z 24.53, 12Z 29.55, 18Z 33.46, 07-09 00Z 29.26, 06Z 26.28                           |
| openweather | Wind direction (wind_direction)              | 07-06 12Z 233.5, 18Z 169.2, 07-07 00Z 178.6, 06Z 178, 12Z 214.6, 18Z 175.7, 07-08 00Z 170.7, 06Z 193.8, 12Z 233.3, 18Z 190.1, 07-09 00Z 178.8, 06Z 172.6                           |
| openweather | Wind speed (wind_speed)                      | 07-06 12Z 1.838, 18Z 2.215, 07-07 00Z 2.186, 06Z 1.73, 12Z 1.439, 18Z 2.646, 07-08 00Z 3.145, 06Z 2.234, 12Z 2.152, 18Z 2.746, 07-09 00Z 2.792, 06Z 1.932                          |

| Source     | Metric                                                | Values                                                                                                                                                           |
|------------|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| purpleair  | PM2.5 (pm25)                                          | 07-06 12Z 8.371, 18Z 6.384, 07-07 00Z 6.413, 06Z 7.879, 12Z 7.973, 18Z 8.328, 07-08 00Z 7.524, 06Z 10, 12Z 7.866, 18Z 5.304, 07-09 00Z 5.023, 06Z 6.842, 12Z 8.4 |
| sentinel5p | Aerosol index granules (s5p_aer_ai_granule_available) | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| sentinel5p | CH4 granules available (s5p_ch4_granule_available)    | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| sentinel5p | CO column (s5p_co_column)                             | 07-06 18Z 0.02528, 07-07 18Z 0.02495, 07-08 18Z 0.02437                                                                                                          |
| sentinel5p | CO granules available (s5p_co_granule_available)      | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| sentinel5p | HCHO column (s5p_hcho_column)                         | 07-06 18Z 0.0002254, 07-07 18Z 0.0002152, 07-08 18Z 0.000185                                                                                                     |
| sentinel5p | HCHO granules available (s5p_hcho_granule_available)  | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| sentinel5p | NO2 column (s5p_no2_column)                           | 07-06 18Z 4.086e-05, 07-07 18Z 3.867e-05, 07-08 18Z 4.307e-05                                                                                                    |
| sentinel5p | NO2 granules available (s5p_no2_granule_available)    | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| sentinel5p | O3 granules available (s5p_o3_granule_available)      | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| sentinel5p | SO2 column (s5p_so2_column)                           | 07-06 18Z 4.559e-05, 07-07 18Z 2.628e-06, 07-08 18Z -2.084e-05                                                                                                   |
| sentinel5p | SO2 granules available (s5p_so2_granule_available)    | 07-06 18Z 1, 07-07 18Z 1, 07-08 18Z 1                                                                                                                            |
| tceq       | Carbon monoxide (co)                                  | 07-06 12Z 0.2353, 18Z 0.1882, 07-07 00Z 0.23, 06Z 0.25, 12Z 0.2167, 18Z 0.2706, 07-08 00Z 0.2, 06Z 0.2444, 12Z 0.25, 07-09 06Z 0.2222, 12Z 0.5                   |
| tceq       | Nitrogen dioxide (no2)                                | 07-06 12Z 5.182, 18Z 4.274, 07-07 00Z 4, 06Z 6.039, 12Z 5.905, 18Z 7.493, 07-08 00Z 6.903, 06Z 10.96, 12Z 7.344, 18Z 1.1, 07-09 06Z 5.981, 12Z 9.275             |
| tceq       | Sulfur dioxide (so2)                                  | 07-06 12Z 0.1353, 18Z 0.075, 07-07 00Z -0.23, 06Z -0.2687, 12Z 0.3333, 18Z -0.05294, 07-08 00Z -0.2, 06Z -0.1867, 12Z -0.03889, 07-09 06Z -0.1556, 12Z -0.1      |

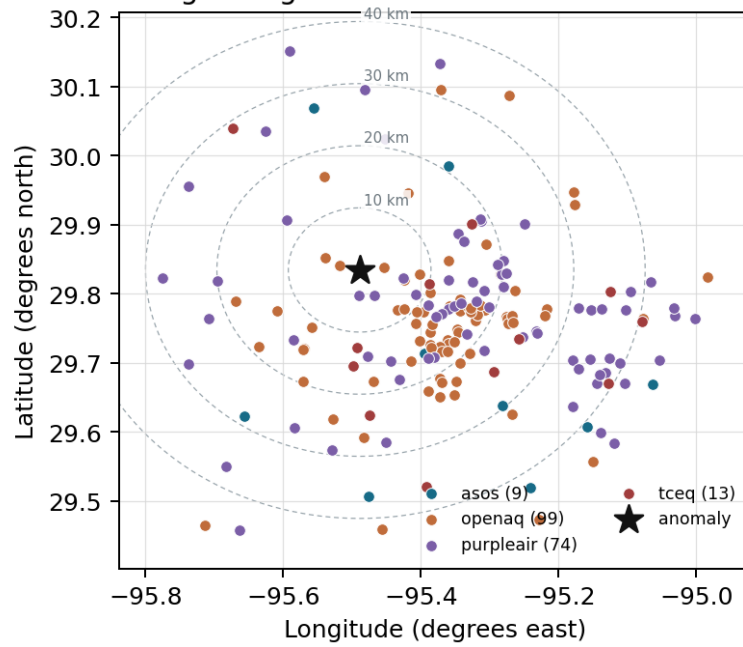
## Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

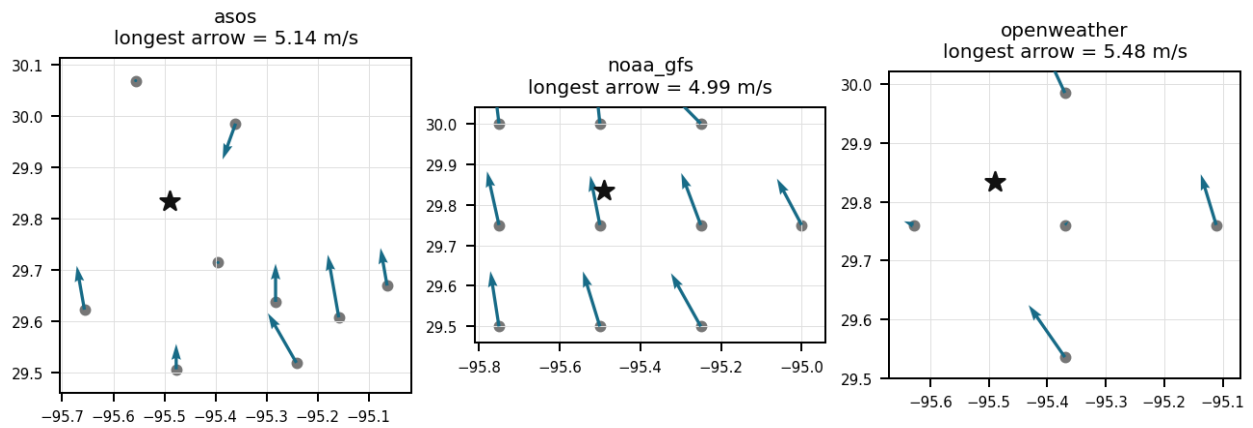
| Source   | Metric                                       | Values                                                                                                                                                                                                                                                                                               |
|----------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asos     | Relative humidity (humidity)                 | MCJ (16.161 km) 72.53, IAH (20.788 km) 72.76, DWH (26.788 km) 79.7, SGR (28.573 km) 76.96, HOU (29.611 km) 72.62, AXH (36.503 km) 73.08, EFD (40.678 km) 68.57, LVJ (42.433 km) 71.95, +1 more                                                                                                       |
| asos     | Air temperature (temperature)                | MCJ (16.161 km) 29.01, IAH (20.788 km) 28.67, DWH (26.788 km) 27.11, SGR (28.573 km) 28.72, HOU (29.611 km) 29.39, AXH (36.503 km) 28.32, EFD (40.678 km) 30.37, LVJ (42.433 km) 29.18, +1 more                                                                                                      |
| asos     | Wind direction (wind_direction)              | MCJ (16.161 km) 142.9, IAH (20.788 km) 157.4, DWH (26.788 km) 85.88, SGR (28.573 km) 152.6, HOU (29.611 km) 165.4, AXH (36.503 km) 78.93, EFD (40.678 km) 145.6, LVJ (42.433 km) 112.8, +1 more                                                                                                      |
| asos     | Wind speed (wind_speed)                      | MCJ (16.161 km) 2.909, IAH (20.788 km) 2.392, DWH (26.788 km) 1.252, SGR (28.573 km) 2.572, HOU (29.611 km) 2.855, AXH (36.503 km) 0.915, EFD (40.678 km) 3.195, LVJ (42.433 km) 1.779, +1 more                                                                                                      |
| noaa_gfs | 500-hPa geopotential height (gh_500)         | gfs:29.75,-95.50 (9.422 km) 5909, gfs:30.00,-95.50 (18.465 km) 5909, gfs:29.75,-95.25 (24.902 km) 5909, gfs:29.75,-95.75 (26.859 km) 5909, gfs:30.00,-95.25 (29.512 km) 5909, gfs:30.00,-95.75 (31.178 km) 5909, gfs:29.50,-95.50 (37.177 km) 5910, gfs:29.50,-95.25 (43.758 km) 5910, +2 more       |
| noaa_gfs | Planetary boundary layer height (pbl_height) | gfs:29.75,-95.50 (9.422 km) 812.8, gfs:30.00,-95.50 (18.465 km) 921.9, gfs:29.75,-95.25 (24.902 km) 821, gfs:29.75,-95.75 (26.859 km) 620.4, gfs:30.00,-95.25 (29.512 km) 897.2, gfs:30.00,-95.75 (31.178 km) 579.5, gfs:29.50,-95.50 (37.177 km) 636.2, gfs:29.50,-95.25 (43.758 km) 588.5, +2 more |

| Source      | Metric                                  | Values                                                                                                                                                                                                                                                                                                           |
|-------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| noaa_gfs    | Precipitable water (precipitable_water) | gfs:29.75,-95.50 (9.422 km) 45.57, gfs:30.00,-95.50 (18.465 km) 45.9, gfs:29.75,-95.25 (24.902 km) 46.51, gfs:29.75,-95.75 (26.859 km) 44.79, gfs:30.00,-95.25 (29.512 km) 47.24, gfs:30.00,-95.75 (31.178 km) 44.89, gfs:29.50,-95.50 (37.177 km) 44.76, gfs:29.50,-95.25 (43.758 km) 45.74, +2 more            |
| noaa_gfs    | Surface pressure (surface_pressure)     | gfs:29.75,-95.50 (9.422 km) 1013, gfs:30.00,-95.50 (18.465 km) 1012, gfs:29.75,-95.25 (24.902 km) 1015, gfs:29.75,-95.75 (26.859 km) 1012, gfs:30.00,-95.25 (29.512 km) 1013, gfs:30.00,-95.75 (31.178 km) 1010, gfs:29.50,-95.50 (37.177 km) 1014, gfs:29.50,-95.25 (43.758 km) 1015, +2 more                   |
| noaa_gfs    | 850-hPa temperature (t_850)             | gfs:29.75,-95.50 (9.422 km) 19.09, gfs:30.00,-95.50 (18.465 km) 19.08, gfs:29.75,-95.25 (24.902 km) 19.03, gfs:29.75,-95.75 (26.859 km) 19.15, gfs:30.00,-95.25 (29.512 km) 19.02, gfs:30.00,-95.75 (31.178 km) 19.21, gfs:29.50,-95.50 (37.177 km) 19.09, gfs:29.50,-95.25 (43.758 km) 19.11, +2 more           |
| noaa_gfs    | 10-m eastward wind (u_10m)              | gfs:29.75,-95.50 (9.422 km) 0.3685, gfs:30.00,-95.50 (18.465 km) 0.6292, gfs:29.75,-95.25 (24.902 km) 0.3454, gfs:29.75,-95.75 (26.859 km) 0.5731, gfs:30.00,-95.25 (29.512 km) -0.09693, gfs:30.00,-95.75 (31.178 km) 0.8569, gfs:29.50,-95.50 (37.177 km) 0.1892, gfs:29.50,-95.25 (43.758 km) 0.2238, +2 more |
| noaa_gfs    | 10-m northward wind (v_10m)             | gfs:29.75,-95.50 (9.422 km) 2.288, gfs:30.00,-95.50 (18.465 km) 2.146, gfs:29.75,-95.25 (24.902 km) 2.182, gfs:29.75,-95.75 (26.859 km) 2.46, gfs:30.00,-95.25 (29.512 km) 1.93, gfs:30.00,-95.75 (31.178 km) 2.434, gfs:29.50,-95.50 (37.177 km) 1.995, gfs:29.50,-95.25 (43.758 km) 1.907, +2 more             |
| openaq      | Ozone (ozone)                           | 319 (0.008 km) 0.0156, 1319333 (11.294 km) 0.02115, 311 (15.421 km) 0.01672, 310 (17.39 km) 0.02088, 287 (18.775 km) 0.01842, 2046 (19.778 km) 0.01744, 312 (23.429 km) 0.01646, 325 (24.945 km) 0.0056, +9 more                                                                                                 |
| openaq      | PM10 (pm10)                             | 15852419 (18.045 km) 27.5, 13380082 (24.981 km) 33, 271 (39.309 km) 16.47                                                                                                                                                                                                                                        |
| openaq      | PM2.5 (pm25)                            | 10762809 (2.905 km) 9.653, 12337668 (3.368 km) 7.042, 14140750 (5.232 km) 6.688, 8967015 (6.377 km) 5.567, 8967363 (8.26 km) 7.353, 13787197 (8.381 km) 7.015, 8967413 (8.848 km) 6.67, 8967373 (10.454 km) 7.197, +71 more                                                                                      |
| openweather | Cloud cover (cloud_cover)               | grid:center (14.139 km) 26.94, grid:west (15.756 km) 22.53, grid:north (20.333 km) 37.36, grid:south (35.126 km) 24.22, grid:east (37.388 km) 23.38                                                                                                                                                              |
| openweather | Relative humidity (humidity)            | grid:center (14.139 km) 76.19, grid:west (15.756 km) 75.65, grid:north (20.333 km) 73.82, grid:south (35.126 km) 74.21, grid:east (37.388 km) 78.79                                                                                                                                                              |
| openweather | Precipitation rate (precipitation)      | grid:center (14.139 km) 0.08208, grid:west (15.756 km) 0.02028, grid:north (20.333 km) 0.01014, grid:south (35.126 km) 0.001944, grid:east (37.388 km) 0.1236                                                                                                                                                    |
| openweather | Surface pressure (pressure)             | grid:center (14.139 km) 1016, grid:west (15.756 km) 1016, grid:north (20.333 km) 1016, grid:south (35.126 km) 1016, grid:east (37.388 km) 1016                                                                                                                                                                   |
| openweather | Air temperature (temperature)           | grid:center (14.139 km) 29.72, grid:west (15.756 km) 29.49, grid:north (20.333 km) 29.47, grid:south (35.126 km) 29.51, grid:east (37.388 km) 29.56                                                                                                                                                              |
| openweather | Wind direction (wind_direction)         | grid:center (14.139 km) 205.3, grid:west (15.756 km) 177.4, grid:north (20.333 km) 173.9, grid:south (35.126 km) 201.8, grid:east (37.388 km) 194.5                                                                                                                                                              |
| openweather | Wind speed (wind_speed)                 | grid:center (14.139 km) 2.053, grid:west (15.756 km) 1.977, grid:north (20.333 km) 2.208, grid:south (35.126 km) 2.16, grid:east (37.388 km) 2.876                                                                                                                                                               |
| purpleair   | PM2.5 (pm25)                            | 51959 (4.098 km) 6.977, 301969 (4.533 km) 10.2, 155367 (6.247 km) 8.124, 223813 (8.787 km) 7.406, 302037 (11.11 km) 9.547, 246011 (12.486 km) 6.386, 130763 (12.96 km) 2.36, 296101 (13.109 km) 7.91, +66 more                                                                                                   |
| tceq        | Carbon monoxide (co)                    | 482011052 (10.018 km) 0.3893, 482011035 (24.981 km) 0.1216, 482011039 (39.322 km) 0.1891                                                                                                                                                                                                                         |
| tceq        | Nitrogen dioxide (no2)                  | 482010047 (0.0 km) 7.938, 482011052 (10.018 km) 13.83, 482011066 (12.524 km) 5.505, 482010055 (15.427 km) 2.99, 482010024 (17.385 km) 6.465, 482010416 (24.952 km) 4.863, 482011035 (24.981 km) 10.48, 482010029 (28.955 km) 2.558, +4 more                                                                      |
| tceq        | Sulfur dioxide (so2)                    | 482010051 (23.42 km) -0.1241, 482011035 (24.981 km) -0.05385, 482011039 (39.322 km) 0.04375                                                                                                                                                                                                                      |

Ground observation locations in the stored 72-hour context  
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



## Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

| Label definitions, from the labeling guide                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.                                                                                            |
| Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.                                                       |
| Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging. |
| Missing or thin evidence means Unsure, not Invalid.                                                                                                                                              |
| A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.                                                                                                 |
| Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.                                                     |

### Claim 1 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note (optional):

### Claim 2 of 36

Surface wind observations from ASOS stations recorded calm conditions reaching 0.0 m/s near the anomaly location around 2026-07-08T00:00:00Z, facilitating localized pollutant accumulation through atmospheric stagnation.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note (optional):

### Claim 3 of 36

Regional smoke transport or a broad sulfur-rich industrial episode is not strongly supported because Sentinel-5P CO and NO<sub>2</sub> columns near Houston were not unusually elevated before the event, Sentinel-5P SO<sub>2</sub> was near zero on 2026-07-07 18Z, TCEQ SO<sub>2</sub> near the event was -0.4 ppb, and area PM<sub>2.5</sub> remained modest at 4.586 ug/m<sup>3</sup> in the 00Z six-hour mean.

Mark exactly one:

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Note (optional):

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### Claim 4 of 36

The TCEQ monitoring network recorded a severe NO<sub>2</sub> concentration anomaly reaching 28.8 ppb at 2026-07-08T00:00:00Z in Houston, Texas.

Mark exactly one:

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Note (optional):

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### Claim 5 of 36

The air quality anomaly in Houston, Texas on July 8th was characterized by elevated levels of PM<sub>2.5</sub>.

Mark exactly one:

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Note (optional):

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### Claim 6 of 36

ASOS meteorological stations near the anomaly site observed calm surface winds with speeds dropping to 0.0 m/s around 2026-07-07T23:55:00Z.

Mark exactly one:

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Note (optional):

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### Claim 7 of 36

The anomalous TCEQ NO2 value of 28.8 ppb exceeds the expected value of 4.731724137931035 ppb by about 24.1 ppb and has a reported z-score of 6.803119928227577 with severe severity.

Mark exactly one:

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Note (optional):

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### Claim 8 of 36

The air quality anomaly in Houston, Texas is not caused by high levels of NO2.

Mark exactly one:

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Note (optional):

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**Claim 9 of 36**

The Houston NO<sub>2</sub> anomaly is consistent with enhanced nighttime accumulation because the nearest ASOS station reported calm winds at 23:55Z, the 06Z six-hour ASOS wind-speed mean was only 0.3517 m/s across the area, and GFS planetary boundary layer height dropped from about 1283 m at 00Z to about 251 m by 06Z, indicating increasingly poor dispersion overnight.

**Mark exactly one:**☐ V☐ I☐ U**Note (optional):**

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**Claim 10 of 36**

Weak near-surface dispersion is a likely contributing cause because the nearest ASOS station reported calm wind at 23:55Z, the 06Z ASOS 6-hour mean wind speed dropped to 0.35 m/s, and very light winds favor local NO<sub>2</sub> buildup near sources in the Houston urban boundary layer.

**Mark exactly one:**☐ V☐ I☐ U**Note (optional):**

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**Claim 11 of 36**

A short-duration plume from nearby combustion sources such as roadway traffic, industrial combustion, or diesel equipment is a likely physical cause because the anomalous TCEQ monitor measured 28.8 ppb NO<sub>2</sub> at 00Z while nearby area-wide TCEQ NO<sub>2</sub> conditions were much lower on average and the closest TCEQ CO monitor showed contemporaneous combustion-related CO present at 0.6 ppm.

**Mark exactly one:**☐ V☐ I☐ U**Note (optional):**

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### Claim 12 of 36

A severe thunderstorm or tornado event in the region has stirred up dust and pollutants, contributing to the air quality anomaly.

Mark exactly one:

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Note (optional):

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### Claim 13 of 36

TCEQ ground monitors detected a sharp increase in surface NO<sub>2</sub> to 28.8 ppb at the anomaly site, aligned with an increase in total NO<sub>2</sub> column density observed by Sentinel-5P over the larger Houston domain.

Mark exactly one:

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Note (optional):

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### Claim 14 of 36

The pollutant anomaly is a TCEQ NO<sub>2</sub> measurement of 28.8 ppb at 2026-07-08T00:00:00+00:00 at coordinates 29.834, -95.489 in Houston.

Mark exactly one:

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Note (optional):

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### Claim 15 of 36

The air quality anomaly is related to PM<sub>2.5</sub>.

Mark exactly one:

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Note (optional):

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**Claim 16 of 36**

The Houston NO<sub>2</sub> anomaly is consistent with a short-lived nearby combustion plume because the anomalous TCEQ monitor measured 28.8 ppb NO<sub>2</sub> at 00Z while the nearest colocated ozone observation was only 0.015 ppm and nearby TCEQ CO reached 0.6 ppm, which is a fresh-emission pattern with titration of ozone by recently emitted NO.

**Mark exactly one:**☐ V☐ I☐ U**Note (optional):**

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**Claim 17 of 36**

Accumulated local evening rush-hour vehicular emissions under calm atmospheric conditions directly contributed to the sharp spike in NO<sub>2</sub> concentrations observed by TCEQ.

**Mark exactly one:**☐ V☐ I☐ U**Note (optional):**

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**Claim 18 of 36**

The TCEQ monitor at 29.834, -95.489 recorded 28.8 ppb NO<sub>2</sub> at 2026-07-08T00:00:00Z, which was far above the monitor-specific expected value of 4.73 ppb and above the contemporaneous 6-hour area mean of 6.90 ppb, indicating a strongly localized NO<sub>2</sub> enhancement.

**Mark exactly one:**☐ V☐ I☐ U**Note (optional):**

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### Claim 19 of 36

A severe nitrogen dioxide (NO<sub>2</sub>) concentration anomaly of 28.8 ppb was detected at coordinates (29.834, -95.489) on 2026-07-08T00:00:00+00:00, significantly exceeding the expected baseline value of 4.73 ppb.

Mark exactly one:

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Note (optional):

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### Claim 20 of 36

NOAA GFS model data shows planetary boundary layer height collapsing from nearly 2000 meters at 18:00 UTC on July 7 to 976 meters by 00:00 UTC on July 8, 2026, trapping ground-level nitrogen dioxide emissions within a shrinking boundary layer.

Mark exactly one:

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Note (optional):

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### Claim 21 of 36

The anomaly occurred on July 8th, 2026, around 00:01:52 UTC, and was located approximately 15.756 km from the query point.

Mark exactly one:

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Note (optional):

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### Claim 22 of 36

The PM2.5 levels are elevated at 18.7 ug/m3, which is higher than the mean of 7.33 ug/m3.

Mark exactly one:

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Note (optional):

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### Claim 23 of 36

The anomaly was likely caused by a combination of factors, including local emissions and weather patterns.

Mark exactly one:

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Note (optional):

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### Claim 24 of 36

Sentinel-5P recorded an elevated tropospheric NO2 column density averaging 4.31e-5 mol/m<sup>2</sup> over the area during the 18Z window on July 8, 2026.

Mark exactly one:

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Note (optional):

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### Claim 25 of 36

The air quality anomaly in Houston, Texas is not caused by high levels of SO2.

Mark exactly one:

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Note (optional):

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### Claim 26 of 36

Fresh local NO<sub>x</sub> emissions with limited immediate oxidation are a likely cause because ozone near the anomaly was only 0.015 ppm at 00Z, TCEQ SO<sub>2</sub> near the time was not elevated, PM<sub>2.5</sub> remained modest across OpenAQ and PurpleAir, and Sentinel-5P did not indicate a coincident regional combustion or sulfur event over Houston.

Mark exactly one:

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Note (optional):

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### Claim 27 of 36

The 6-hour mean across 12 TCEQ NO<sub>2</sub> monitors for 2026-07-08 00Z is 6.903 ppb, which is far below the anomalous monitor reading of 28.8 ppb at the same time.

Mark exactly one:

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Note (optional):

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### Claim 28 of 36

NO<sub>2</sub> levels were also significantly higher than normal during the same period.

Mark exactly one:

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Note (optional):

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### Claim 29 of 36

NOAA GFS model output showed the planetary boundary layer height rapidly dropping from nearly 2000 meters at 18:00 UTC down to approximately 976 meters at 00:00 UTC on July 8.

Mark exactly one:

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Note (optional):

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### Claim 30 of 36

Near-zero surface wind speeds measured by ASOS stations hindered horizontal ventilation, preventing the dispersion of emitted nitrogen dioxide near the TCEQ station.

Mark exactly one:

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Note (optional):

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### Claim 31 of 36

OpenAQ ozone near the anomaly location was 0.015 ppm at 2026-07-08T00:00:00Z, which is consistent with a fresh NO<sub>x</sub> plume that had not produced elevated ozone and may have locally suppressed ozone through titration.

Mark exactly one:

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Note (optional):

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### Claim 32 of 36

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to the accumulation of pollutants in the area.

Mark exactly one:

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Note (optional):

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### Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:

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Note (optional):

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### Claim 34 of 36

The collapse of the planetary boundary layer height depicted in GFS reanalysis constrained vertical mixing, trapping urban NOx emissions close to the surface.

Mark exactly one:

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Note (optional):

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### Claim 35 of 36

Surface wind speed measured near the anomaly location dropped to 0.0 m/s at 2026-07-07T23:55:00+00:00.

Mark exactly one:

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Note (optional):

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Claim 36 of 36

ASOS observations nearest the anomaly showed 0.0 m/s wind at 2026-07-07T23:55:00Z and the 06Z period had a 6-hour mean wind speed of only 0.35 m/s, supporting weak dispersion that would trap a nearby combustion plume near the surface.

Calm-wind flag: wind direction unstable under calm conditions. openweather: event wind 0.89 m/s, below its cutoff of 1.5 m/s (mean - 2\*SD gave -0.38737025451646634 m/s; n=360 wind readings). Cutoff floor: confirmed 2026-07-24.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note (optional):

**Optional: most likely cause**

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

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